College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 03

Focus: Classes, encapsulation, constructors intro, simple validation

Difficulty: Easy-Moderate

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Class anatomy.

a. Label each of the following as field, method, or neither: `speed`, `accelerate()`, `package`, `brand`. **(4 marks)**

b. Write a `Vehicle` class with fields `brand` and `speed`, plus methods `accelerate()` (adds 10 to speed) and `display()`. **(10 marks)**

c. Create two vehicles in `main` with different brands and show their speeds after one accelerate each. **(6 marks)**

Question 2 [20 Marks]

Access modifiers.

a. Explain `public` vs `private` with one example each. **(6 marks)**

b. What compile-time problem occurs if `main` tries to write to a private field directly? **(4 marks)**

c. Refactor a public field `balance` into private + getter + deposit method with a positive-amount check. **(10 marks)**

Question 3 [20 Marks]

```
public class BankAccount {
    private double balance;
    public void deposit(double amount) {
        if (amount > 0) balance += amount;
    }
}
```

Study the fragment.

a. Why can `deposit` change `balance` even though the field is private? **(4 marks)**

b. Add `withdraw(double amount)` that refuses negative amounts and amounts greater than balance. **(8 marks)**

c. Add `getBalance()` and show three calls from `main`: deposit, withdraw, print. **(8 marks)**

Question 4 [20 Marks]

Constructors - introduction.

a. What is a constructor? How does its name relate to the class name? **(4 marks)**

b. Does a constructor have a return type (including `void`)? Explain. **(4 marks)**

c. Write a default constructor and a parameterized constructor for class `Product` with fields `name` and `price`. **(8 marks)**

d. When does a constructor run? **(4 marks)**

Question 5 [20 Marks]

The keyword `this` (simple).

a. What does `this` refer to inside an instance method? **(4 marks)**

b. Write `setPrice(double price) { this.price = price; }` and explain each `price`. **(8 marks)**

c. What problem appears if you write `price = price;` inside that setter? **(4 marks)**

d. Give one other situation where `this` is useful. **(4 marks)**

Question 6 [20 Marks]

Rectangle: procedural vs object (bridge).

a. Describe how a procedural Rectangle program stores width and height. **(4 marks)**

b. Write an OO `Rectangle` with private fields, a constructor, `area()`, and `perimeter()`. **(10 marks)**

c. State two benefits of the OO version for a program that needs many rectangles. **(6 marks)**

Question 7 [20 Marks]

Library Phase 1 style - Book.

a. List suitable fields for a library `Book` (at least four). **(4 marks)**

b. Write `borrow()` and `returnBook()` methods that prevent double-borrow / double-return using a boolean flag. **(10 marks)**

c. Explain why availability should not be set freely from outside without rules. **(6 marks)**

******* END OF QUESTION PAPER *******